Response to the reviewers

Ronald Rethabile Malamule Mahomane
Department of Computing, Mathematical and Statistical Sciences, University of Namibia, Windhoek
School of Pathology, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg
Basis Energy, Victoria, Australia
ron@basisenergygroup.com

September 2026

Scope

This letter answers a pre-submission review of two manuscripts by a panel of seven referees: sequential statistics, statistical disclosure control, differential privacy, central bank statistics, machine learning, reproducibility, and a general referee who took the part of Reviewer 2. The manuscripts are

- *Adaptive certification of attack-conditional disclosure risk across release programmes*, with its supplement, for the Journal of Privacy and Confidentiality (the article); and
- *Governing privacy risk across repeated statistical releases: probability models, programme tolerances and accountable decisions*, a long-form author manuscript for IFC Working Paper consideration (the long form).

Titles changed in revision; the first-round reports use the old ones. The full reports, including a second-round check by Reviewer 2, are in `docs/panel_review/` of the repository.

Summary of changes

Four changes alter what the manuscripts claim.

1. **The pooled credit-register excess is split into its two causes.** In the original attack the pooled adversary held one exposure reading per vintage, so it held six times the auxiliary information of the single-release adversary. A new table (long form, Table 12; article, Table 3) changes one factor at a time on the same targets. With no suppression the rate is 0.079 for one release and one reading and 0.209 for the full pooled attack. Extra readings alone give 0.132; pooling the six vintages with one reading gives 0.136, with an upper bound of 0.145, above the tolerance of 0.12. The programme finding survives, but both manuscripts now attribute only part of the gap to pooling.
2. **The cell-count floor is chosen by the upper bound, on a finer grid.** The old headline chose $k_{\min} = 50$ by point estimate, and 50 was the top of the grid. The revised run adds a fine grid (20 to 100) and applies the exact upper bound (long form, Table 13). The smallest tested floor that meets the tolerance is now 40, which keeps 23,068 loans against 32,581 with no floor.

3. **Noisy counts are composed across vintages.** The pooled column for the noisy-count policies had repeated the single-release value. Each vintage now draws fresh noise and the adversary combines the readings, so the programme spends six times the stated ε. The count rows measure class uniqueness, not loan identification, and both manuscripts now say they cannot be compared with the loan tolerance.

4. **The article runs its gate on a disclosure problem.** A new experiment (`scripts/adaptive_audit_credit.py`, article Table 2) registers a single-release and a pooled coordinate for five policies on the synthetic credit register, over 20 seeds, and compares the gate with a frozen fixed-sample family at the same maximum budget. Every policy passes its single-release coordinate. Only $k = 50$ passes the pooled coordinate with high probability (19 of 20 seeds). At equal budget the frozen family certifies more often; the gate stops earlier.

The code changes behind these results, and the repository fixes, are listed at the end.

Reviewer 2

R2.1 (major). *Theorem 1 is online alpha-spending over test supermartingales, and the relevant literature is not cited.*

Response. Agreed. The article has a new related-work section (Section 2) that cites alpha-investing (Foster and Stine 2008), online familywise error control (Tian and Ramdas 2021, Statistical Methods in Medical Research), test martingales and e-values (Shafer et al. 2011; Vovk and Wang 2021; Ramdas et al. 2023), betting confidence sequences (Howard et al. 2021; Waudby-Smith and Ramdas 2024) and Ville (1939). A remark after the theorem states the reduction to online Bonferroni and names what the article adds: the coordinate definition, the attack-sampling condition, the pooled-coordinate requirement, the audit-cost rule and the measured application.

R2.2 (major). *The paper hands the novelty judgement to the referee.*

Response. Every such sentence is deleted, including those on priority, "the journal mathematics" and the author's status. The abstract ends with a statement of what the gate requires and costs.

R2.3 (major). *Proposition 1 is an intersection attack; Ganta et al. is not cited; the construction needs a shared identifier; q has three meanings.*

Response. The article states a formal threat model (Assumption 1) in which each release carries a pseudonym constant across vintages and the adversary knows the target's full quasi-identifier vector. The proposition uses the long form's construction, with s for the alphabet size, and is described as a standard non-composition example that fixes the estimand, citing Sweeney (2002), Wang and Fung (2006), Ganta, Kasiviswanathan and Smith (2008) and Dinur and Nissim (2003). The long form now states the same pseudonym condition in its proposition and in the credit-register threat model, and notes that a rotating pseudonym would make the pooled figures overstate the risk.

R2.4 (major). *The gate is never run on a disclosure problem.*

Response. Partly addressed. The new credit-register experiment above runs the gate with adaptive registration and compares it with a frozen family at equal budget. Not done: a run on the public Census file, comparison with the beta-binomial confidence sequence and with an adaptive-bet e-process, and a case with reused targets. The article says so. The experiment also

does not exercise adaptive registration in a way a frozen family could not match, because every single-release coordinate certified in every seed; the article states this.

R2.5 (major). *The overlap with the long form is not disclosed in the manuscript.*

Response. The long form is cited as a reference. A first-page footnote names the results reproduced from it and the results that are new (the theorem, the weights and cost propositions, the toy audit and the credit-register audit). The carried-over tables are cut to the decomposition table the argument needs, and the supplement repeats the statement. The cover letter will repeat it again and mention the Lisbon presentation.

R2.6 (major). *The neural section does not belong in the article.*

Response. The main text now has one paragraph: any predictor may supply the bet if it is fixed before the outcome is seen, and validity does not depend on its accuracy. The network, its gradient, the Lipschitz bound (now with the input-standardisation factor) and the benchmark are in the supplement. The benchmark was rerun to convergence over ten seeds on a third held-out set: logistic regression has test log loss 0.2556 and the three-layer network 0.2576, so the old claim that depth helps is withdrawn.

R2.7 (major). *Replacing α by α/q after inspection is invalid.*

Response. Corrected in the long form: α/q applies to a family fixed before any outcome is seen; hypotheses added after inspection need a registered spending sequence, as in the article.

R2.8 (major). *The long form is too long and lists ten contributions.*

Response. Partly addressed. The contribution list has four items, the unsupported and overstated sentences are reworded, the four non-central-bank case studies and the neural material are in appendices, and a reader's guide sends a central bank reader to the governance rule, the credit evaluation and the protocol. The full reordering was not done. The paper is 74 pages, because the decision record, two new tables and the added caveats outweigh what moved.

R2.9 (major). *The article is not organised as a journal article.*

Response. It now runs: introduction with contributions, related work, setting and threat model, gate and theory, experiments, discussion and limitations, conclusion. Terms are defined before use.

R2.10 (minor). *References are thin or incomplete.*

Response. The article's list grew from 8 to 56 entries and the supplement now has its own list of 16; the long form grew from 61 to 109. Each added work is cited where it supports a sentence, and each entry was checked against its Crossref, OpenAlex or publisher record. Bates et al. is cited as J. ACM 68(6); Xu et al. as NeurIPS 2024; Fellegi–Sunter is no longer cited for attacker models (Duncan and Lambert 1989 is). Zhou et al. (arXiv:2605.07002, May 2026) was checked on arXiv by the differential privacy referee and is kept. Still open: Elliot and Dale (1999) is left out of the article until its record is confirmed; the long form lacks a report number for Duncan et al. (2001) and does not yet cite Lambert (1993). The long-form bibliography is alphabetical by first author.

R2.11 (minor). *Affiliations and corresponding author.*

Response. Both article files carry the full affiliation block and the corresponding author's email.

R2.12 (minor). *Notation.*

Response. Fixed as requested, with further changes after the second round: the Clopper–Pearson bound is $\overline{R}(Y, B; \lambda)$, so U is only the population, K_v only the key and γ only a stopping

time; α is defined where first used; the abstract says "at least its tolerance".

R2.13 (minor). *Unclear table entries.*

Response. "Unique share" is dropped with its table. The $k = 5$ pooled figure is explained as Monte Carlo variation of 58 in 4,000 targets that reverses on a second draw. The decision illustration uses the upper bound.

R2.14 (minor). *Monte Carlo standard errors in the toy audit.*

Response. Added. The overall false-certification rate is reported against α with its standard error, and the adaptive-registration run is described as another estimate of the single-coordinate rate, since the streams are independent.

R2.15 (minor). *Unlabelled and uncited figures.*

Response. The pooling figure is labelled and cited; the SENTRY-R diagram moved to the supplement; the composition figure was dropped.

R2.16 (minor). *Undefined terms, puffery, meta-commentary.*

Response. SENTRY-R and LLM are defined at first use. The listed phrases are removed from both manuscripts, together with the further stock phrases the second round found. The network output is no longer called "trust"; the quantity $1 - \overline{R}_i/\tau_i$ is the audited margin.

R2.17 (minor). *The two titles give no sign the papers are companions.*

Response. Both titles now refer to repeated releases or release programmes, and each manuscript names the other.

Second round

Reviewer 2's second-round check found ten further problems, all now fixed: the article had merged the coarse and fine grids and read low audit power at $k = 40$ as evidence of excess risk; it claimed an advantage for adaptive registration that its experiment does not test; "about half and half" omitted an interaction of 0.020; the long form called noisy counts admissible while its own caption says they measure a different event; the decision record lacked a caveat on selection across floors and named the wrong vintage; a two-release pooled figure was compared only with the four-release bound (it is 2.8 times the bound of 0.148 for its own two releases); the self-citation gave no route to the manuscript; Foster and Stine was cited for alpha-spending; and the superseded short note is now marked as such in its source. Two reported problems were not errors: the long-form bibliography is in order (entries with one first author sort by co-author), and the toy audit's two runs of 10 in 2,000 use separate random draws.

Other referees: main points

Ref.	Point	Outcome
1.1	The anytime certification cost (12,417 against 2,995) is the price of a uniform mixture, not of monitoring.	Corrected in the long form's text and in the generated table caption.

Ref.	Point	Outcome
1.2–1.4	Certification should hold at first crossing; the budget can be off by one; registered weights can be changed.	Code fixed: the coordinate stops at first crossing, its specification is immutable, each outcome is logged with its bet, and the budget is checked against the crossing trial. New tests.
1.5–1.8	Measurability wording, predictable coordinate index, relaxed-denominator bet range, supplement wording.	Fixed in the article and supplement.
2.1–2.3	Stale falsification figures; wrong census row; credit record breaks the paper's own rule.	Fixed.
2.4, 2.5	The census adversary reads the key exactly; the journalist figure rests on an assumed sampling fraction.	Captions and text corrected; $\hat{\rho}$ reported for fractions from 0.05 (0.225) to 0.001 (0.0016).
2.6, 2.7	Pooled gap from extra readings; counts not composed.	New decomposition, composed counts, bound-based choice (see summary).
2.9–2.12	k-anonymity overreach; journalist measure misnamed; dispersion drift; contributor rules.	Reworded; ρ renamed and El Emam and Dankar cited; drift noted; dominance and p% rules named and secondary suppression labelled heuristic. Not done: $E[1/F \mid f = 1]$ is not reported.
2.15	Notation clashes in the long form.	Not done.
3.1, 3.2	Composition prior art; shared identifier.	Fixed in both manuscripts.
3.3, 3.4	DP with $\delta > 0$; overclaim about formal privacy.	Fixed.
3.5, 3.6	Position against privacy auditing; what JPC needs for novelty.	Auditing literature cited and the asymmetry stated. Not done: comparison with Zhou et al. and Xu et al. at equal budget.
4.1–4.3	Abstract without a recommendation; point-estimate headline; two runs mixed.	New abstract; bound-based choice; separate draws stated once.
4.4, 4.5	Title and neural material; length and order.	Retitled and moved to appendices; full reorder not done.
4.6–4.8	No filled decision record; practice claims unsourced; synthetic caveat.	Decision record (Table 15) with a programme horizon; AnaCredit wording checked against Regulation (EU) 2016/867; an unverifiable Bank of England citation removed; caveats added.
5.1–5.4	SENTRY-R audit not frozen; undertrained baseline; LLM supplies its own provenance; owner and expiry missing.	Code fixed: alpha, planned trials, owner, expiry and development evidence are frozen and checked; an optional ledger refuses a second audit; provenance is set by the caller. Benchmark rerun.
5.10–5.12	Separation only documented; depth unsupported; the word "trust".	Text now says what the code enforces and what it does not. Module names kept for compatibility.

Ref.	Point	Outcome
6.1–6.3	Stale manifest; missing CI; weak number checker.	Manifest regenerated and tested; workflows added; checker matches whole numbers and has a mode for the article's quoted figures.
6.4–6.12	Stale section numbers, non-reproducible figures, NaN in JSON, CLI errors.	Fixed.

Code and repository

All 146 tests pass and `ruff` is clean. The main changes: the missing `colab_setup.py` was restored, so the test suite runs; `evaluation.py` adds the fixed-readings attack, composed noisy counts and the bound-based choice; `adaptive_audit.py` and `sentry_r.py` carry the fixes above; `reporting.py` writes strict JSON and figures without creation dates, and two new tables; `check_paper_numbers.py` has the stricter full mode and the new quoted mode; `rebuild_evidence.py` can publish tables to `paper/generated`; `.github/workflows` holds the CI and refresh workflows the README described. The paper run in `artifacts-paper/` was regenerated; apart from the new results, every number is unchanged except the transport-bound figures for the admissible count release (0.004 and 0.027), which follow from composing the counts.